Ammon Perkes, Ph.D. @alltheperkes.bsky.social

Research Experience

Project Scientist; Postdoctoral Researcher, Kate Laskowski Lab January, 2022 – Present

- Investigated the mechanisms driving dominance hierarchies and the winner effect
- Developed and maintained automated tracking system for fish behavior
- Mentored graduate, undergraduate, and high school students
- Managed and maintained fish facility

Postdoctoral Researcher, Marc Schmidt Lab September, 2021 – December, 2021

- Investigated the importance of collective courtship for reproductive fitness
- Developed automated tracking systems for songbird behavior

Education

Ph.D. in Biology; University of Pennsylvania 2014 – 2021

- Advisor: Marc Schmidt, Ph.D.

B.S. in Biology; Brigham Young University 2011 – 2014

Honors, Awards, Fellowships

NSF Postdoctoral Research Fellowship in Biology 2021 – 2023
Computational Approaches… (CANAC) Training Grant 2018 – 2021
Behavioral and Cognitive Neuroscience (BCN) Training Grant 2017 – 2018
Integrative Graduate Education and Research Traineeship (IGERT) 2015 – 2017

Selected Publications

Gallagher, J. H., **Perkes, A.**, … & Laskowski, K. Born this way: individuality is seeded before birth and robust to environmental stress. *EcoEvoRxiv* 10.32942/X2ZH27

Perkes, A., Laskowski, K.L. "Bayesian updating for self-assessment explains social dominance and winner–loser effects." *Animal Behaviour* 224 (2025): 123191.

Xiao, S., Wang, Y., **Perkes, A**., … & Badger, M. (2023). Multi-view tracking, re-id, and social network analysis of a flock of visually similar birds in an outdoor aviary. *International Journal of Computer Vision*, 131(6), 1532-1549.

Perkes, A., Anderson, H., Schmidt, M., White, D. Male coordination of courtship predicts group-level reproductive output in a songbird. bioRxiv 2021.11.20.469403

Perkes, A., Pfrommer, B., Daniilidis, K., White, D., Schmidt, M. Variation in female songbird state determines signal strength needed to evoke copulation. bioRxiv 2021.05.19.444794

Anderson, H., **Perkes, A**., Gottfried, J., Davies, H., White, D., Schmidt, M. (2021). Female signal jamming in a socially monogamous brood parasite. *Animal Behaviour, 172, 155-169*

Badger, M., Wang, Y., Modh, A., **Perkes, A**., … & Daniilidis, K., (2020). 3D Bird Reconstruction: a Dataset, Model, and Shape Recovery from a Single View. *ECCV*, 2020 pp 1-17.

Perkes, A., White, D., Wild, M. and Schmidt, M., (2019). Female Songbirds: The unsung drivers of courtship behavior and its neural substrates. *Behavioral processes*, 163, 60-70.

Talks

Perkes, A. Winners, losers, and the development of group phenotypes. *ABGG Seminar, UC Davis, 2025, Davis, CA*

Perkes, A. Win, lose, draw inference: Bayes formula predicts…winner-loser effects. *Animal Behavior, 2025, Baltimore, MD*

Perkes, A. Clonal groups show individual and group-level differences in collective behavior. *ISBE, 2024, Melbourne, Australia*

Perkes, A. Clonal fish show individual and group-level differences in collective behavior. *Animal Behavior, 2024, London, ON, Canada*

Perkes, A. The Bayesian winner effect. *Math-bio Seminar, UC Davis, 2023, Davis, CA*

Perkes, A. The Bayesian winner effect over time. *Animal Behavior, 2023, Portland, OR*

Perkes, A. Lessons from cowbirds; the hidden forms and functions of courtship. *ABGG Seminar, UC Davis, 2022, Davis, CA*

Perkes, A. Subtle sociality; group interactions and female responses guide courtship in the brown-headed cowbird. *CPB Seminar, UC Davis, 2022, Davis, CA*

Perkes, A. Male coordination of courtship predicts group-level reproductive output in a songbird. *Animal Behavior, 2021, online*.

Perkes, A. Quantification of mating behavior in songbirds. *Animal Behavior, 2020, online.*

Perkes, A. Fine-grain computer vision tracking of songbird social-interaction throughout the breeding season. *Animal Behavior, 2019, Chicago, IL.*

Perkes, A. Signal Jamming: you can duet (the role of female songbirds in the formation of social networks). *CRPPY GREBE Student Conference, 2019, Princeton, NJ.*

Perkes, A. Computing Courtship: the neural mechanisms of mating posture and female choice. *Penn Neuroscience Retreat, 2018, Philadelphia, PA.*

Perkes, A. The theory and practice of monogamy. *Penn, Princeton, Rutgers, Columbia Phd Student Conference, 2018, Rutgers, NJ.*

Perkes, A. Mating outside the box: aviary data and modeling. *Delaware Valley Avian Courtship Workshop, 2018, Princeton, NJ*

Perkes, A. The un-sung heroines: quantifying female behavior and their impact on social networks in songbirds. *Penn, Princeton, Rutgers, Columbia Phd Student Conference, 2017, Columbia, NY.*

Perkes, A. The unsung heroines: female behavior drives network stability in songbirds. *Emergent Properties of Individual Behavior Workshop, 2017, Lexington, KY.*

Abstracts

Perkes, A., Laskowski, K. Bayesian updating predicts social dominance formation and the winner effect. *Animal Behavior 2022 Abstracts. San Jose, Costa Rica.* Online

Perkes, A., Kolotouros, N., Pfrommer, B., Schmidt, M.. Computing Courtship: Song circuit modulation of mating posture in a female songbird. Submitted. *Neuroscience 2019 Abstracts. Chicago, IL:* Society for Neuroscience, 2019. Online.

Perkes, A., Messier, C., Wild, M., Pfrommer, B., Schmidt, M. Mating posture and female choice in songbirds. Submitted. *Neuroscience 2018 Abstracts. San Diego, CA:* Society for Neuroscience, 2018. Online.

Perkes, A., Jean, I., Pfrommer, B., Schmidt, M., Copulatory display and female choice in songbirds: II. Behavioral quantification. Submitted. *Animal Behavior 2018 Abstracts. Milwaukee, WI*: Animal Behavior Society, 2018. Online.

Perkes, A., Messier, C., Ippolito, D., Wild, M., Schmidt, M. Characterization of mating posture and modulatory nuclei in a female songbird. Program No. 235.16. *Neuroscience 2017 Abstracts. Washington, DC*: Society for Neuroscience, 2017. Online.

Perkes, A., Schmidt, M. The mindful songbird: respiratory feedback in the avian song system. *University of Pennsylvania Biology Retreat 2016, Philadelphia, PA.*

Teaching

Workshop: Computer Vision for Biologists Workshop, UC Davis (Organizer) — 2025

Workshop: Computer Vision for Animal Behavior, Laskowski Lab (Organizer) — 2022

Guest Lecture: Collective Motion (BIOL231: Evolution of Animal Behavior) — Fall, 2019

Teaching Assistant for BIOL: 437 Intro to Computational Biology — 2017 – 2018

- Gave office hours and limited lectures for Junhyong Kim, PhD

Teaching Assistant for Bio 100: Principles of Biology — Fall, 2013

- Gave office hours and graded student work for Riley Nelson, PhD

Teacher, Portuguese — 2011 – 2013

- Taught Portuguese and trained other teachers at Provo Missionary Training Center